

The Topological Engineer’s Handbook: Benevolent High-Energy Events in the OEU Architecture

Version 2.0 — Revised & Scientifically Deepened | June 2026

Classification: Tardigradia Game Engine | Physics Deep Dive Series | TGPU v2.0 **Particle Class:** Cross-system document: Shock events, vacuum decay, pair production, extreme particle states
Source: Extends v1.0OEU_ Benevolent Particle Overpressure.pdf (V1.0)

Δ Change Log V1.0 → V2.0

Parameter	V1.0	V2.0	Source
Schwinger pair production threshold	Schwinger E_{crit} mentioned	V2.0: $E_{Schwinger} = m_e^2 c^3 / (e \hbar) = 1.32 \times 10^{18}$ V/m. First approaches to this field via LUXE (ELI-NP, 2024): 10^{23} W/cm ² laser + electrons at 16.5 GeV achieves $E_{field}/E_S \sim 0.05$. Strong-field QED now testable regime	LUXE 2024; ELI-NP 2024
Vacuum bubble nucleation	False vacuum mentioned	V2.0: Metastable vacuum lifetime lower bound $> 10^{600}$ yr (Di Luzio 2023 improved calculation). Top quark mass critical value $m_t^{critical} = 171.7$ GeV. Measured: 172.5 GeV — universe in metastable phase. V2.0: adds VACUUM_BUBBLE event type as ultra-rare catastrophic event (probability 0 in current epoch)	Di Luzio et al. 2023; Degrassi 2022

Parameter	V1.0	V2.0	Source
Unruh effect temperature	Unruh radiation mentioned	V2.0: Unruh temperature $T_U = \hbar a / (2\pi c k_B)$. At $a = 10^{26} \text{ m/s}^2$: $T_U = 1 \text{ eV}$. First laboratory evidence (2023 analog system): thermal photon distribution matches Unruh prediction in accelerated atomic system. V2.0: particles under extreme acceleration emit Unruh radiation — tracked as VIRTUAL_PHOTON events	Chen & Tajima 1999; 2023 analog confirmation
Hawking point in CMB	Not covered	V2.0: CCC (Conformal Cyclic Cosmology) Hawking point signals: Penrose et al. 2024 update — 5 candidate Hawking radiation circles identified in Planck data. Benevolent overpressure from previous aeon. V2.0: adds CCC_SIGNAL particle type as background field	Penrose et al. 2024; Planck CMB

1. The One-Particle Universe Framework — V2.0 Update

The foundational architecture established in V1.0 — Worldline Monism, the “Origami Universe” metaphor, and the Object-Oriented ontology distinguishing intrinsic from extrinsic properties — is **fully preserved** and remains physically rigorous. V2.0 layers NEW discoveries from 2022-2026 atop this framework.

1.1 WebGPU Compute Architecture (V2.0 Core Infrastructure)

V1.0 designed for WebGL/Three.js on the CPU-side particle update loop. V2.0 upgrades to WebGPU (available in Chrome 113+ May 2023, Firefox 2024):

```
// WGSL Compute Shader - V2.0 Particle Integration
@binding(0) @group(0) var<storage, read_write> particles: array<ParticleState>;

struct ParticleState {
    position: vec4f,
```

```

momentum: vec4f,
spin: vec2f,           // Spinor components
berry_phase: f32,      // Geometric phase accumulated
proper_time: f32,      // tau - worldline parameter
color_charge: u32,     // Packed color in 2 bits (R=0,G=1,B=2)
quantum_numbers: u32,  // Packed: isospin, strangeness, charm, beauty
decay_lifetime: f32,
entanglement_id: i32, // -1 = unentangled; >=0 = entanglement group
}

@compute @workgroup_size(64)
fn updateParticles(@builtin(global_invocation_id) id: vec3u) {
    let i = id.x;
    if (i >= arrayLength(&particles)) { return; }

    var p = particles[i];

    // Feynman-Schwinger proper time evolution
    let dtau = uniforms.dt / p.mass; // proper time step

    // Update worldline parameter
    p.proper_time += dtau;

    // Berry phase accumulation (topological phase)
    p.berry_phase += dot(uniforms.gauge_field, p.momentum.xyz) * dtau;

    // Lorentz force (for charged particles)
    let F = cross(uniforms.B_field, p.momentum.xyz) * p.charge_sign;
    p.momentum.xyz += F * uniforms.dt;
    p.position.xyz += p.momentum.xyz * uniforms.dt / p.energy;

    particles[i] = p;
}

```

1.2 Updated Physical Constants (CODATA 2022)

Constant	V1.0 Value	V2.0 Value (CODATA 2022)
m _e	9.10938356×10 ⁻³¹ kg	9.1093837139×10 ⁻³¹ kg
e	1.60217663×10 ⁻¹⁹ C	1.602176634×10 ⁻¹⁹ C (exact)
ħ	1.054571817×10 ⁻³⁴ J·s	1.054571817×10 ⁻³⁴ J·s (exact)
c	2.99792458×10 ⁸ m/s	2.99792458×10 ⁸ m/s (exact)
α	1/137.036	1/137.035999084 ±0.000000021

2. V2.0 Enhanced Data Structures

Extending V1.0 struct definitions with 2022-2026 discoveries:

```

// V2.0 Universal Particle State – extends V1.0
struct ParticleInstanceV2 {
    // Inherited from V1.0
}

```

```

double proper_time;           // tau worldline parameter (meter)
glm::dvec3 position;          // 3-space vector
glm::dvec3 momentum;          // 3-momentum (relativistic)
float spin_z;                  // +0.5 or -0.5 (or eigenvalue)
float berry_phase;             // Accumulated Berry/Aharonov-Bohm phase

// V2.0 ADDITIONS
float entanglement_r;          // Reduced density matrix purity 0-1
int entanglement_partner_id;    // -1 = none; global particle ID
float flavor_oscillation_phase; // For neutrinos/kaons
uint32_t color_bits;           // 3-bit: R(b0),G(b1),B(b2)
float decay_probability_dt;     // P(decay in dt) for unstable particles
float vacuum_alignment;         // Order parameter for condensate mode

// WebGPU Buffer packing
// Total: ~100 bytes/particle * 10^6 particles = 100 MB GPU memory
float _pad[2];                 // Align to 16 bytes for WGSL
};

```

8. V2.0 Benevolence Metric Update

The “Benevolence” concept central to all OEU documents is given a precise V2.0 mathematical definition grounded in 2022-2026 quantum information theory:

Benevolence = Quantum Coherence + Information Preservation + Symmetry Maintenance

$$B(t) = C(t) \cdot F(t) \cdot S(t)$$

Where: - $C(t)$ = l_1 -norm coherence of particle density matrix (Baumgratz et al. 2014) - $F(t)$ = quantum Fisher information / classical Fisher information (ratio ≥ 1) - $S(t)$ = fraction of symmetries unbroken at time t

V2.0 game mechanic: Benevolence score increases when: - Particles maintain long entanglement coherence (high C) - Topological protections are active (Berry phase winding conserved) - Symmetry groups are preserved in interactions (no spurious symmetry breaking)

*End of v1.0 OEU_ Benevolent Particle Overpressure V2.0 | Tardigradia Game Engine SubParticles Series
This document supersedes V1.0 in all physics specifications.*